

ARCHAEOLOGY

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when archaeologists were still uncovering treasures of the past. In 1911 Hiram Bingham introduced Machu Picchu in the Andes to the scientific world, though local farmers had always known the ruins. In the 1920s Leonard Woolley excavated the "royal tombs" at Ur and found in them lyres with a bull's head. In 1933 the teacher Walenty Szwajcer discovered at Biskupin an Iron Age fortified settlement built around 740 BC – roughly 2673 years earlier. Its wood survived because it was submerged in water and peat. In 1939 at Sutton Hoo in England archaeologists found a burial in a ship, with a helmet and treasures. In 1940 teenagers looking for their dog stumbled into the Lascaux cave. During the war, monuments were saved by specialists known as the "Monuments Men," and in 1947 a young shepherd looking for a goat discovered the Dead Sea Scrolls. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏹 In the 1920s, Leonard Woolley excavated the "royal tombs" at Ur and found musical instruments in them, including...
 - a) pianos
 - b) lyres with a bull's head
 - c) electric guitars
 - d) pipe organs
2. 👁 The most famous discovery of interwar Polish archaeology is an Iron Age fortified settlement at...
 - a) Biskupin
 - b) Gniezno
 - c) Krzemionki
 - d) Wieliczka
3. 🗨 The teenagers who discovered the Lascaux cave in 1940 were looking for...
 - a) their dog
 - b) bats
 - c) a water source
 - d) Templar treasure
4. 🏰 The Biskupin settlement was discovered in 1933 and built around 740 BC. Roughly how many years passed between its construction and its discovery? (Ignore the absence of a year 0.)

5. 🌳 The wooden structures of the Biskupin settlement survived about 2,700 years because...
 - a) they were covered with paint
 - b) they were submerged in water and peat
 - c) desert sand covered them
 - d) they froze
6. 🧑 At Sutton Hoo in England (1939), archaeologists found a burial in...
 - a) catacombs
 - b) a ship, with a helmet and treasures
 - c) a pyramid
 - d) a cave with paintings
7. 💖 The Dead Sea Scrolls were discovered in 1947 by...
 - a) a young shepherd looking for a goat
 - b) an army general
 - c) a diver
 - d) a tourist with a camera
8. 🌿 Machu Picchu in the Andes was introduced to the scientific world in 1911 by...
 - a) Howard Carter
 - b) Roald Amundsen
 - c) Indiana Jones
 - d) Hiram Bingham
9. ⭐ During World War II, many monuments and artworks were saved by specialists known as the...
 - a) "Musketeers"
 - b) "Monuments Men"
 - c) "Ghostbusters"
 - d) "Templars"

PHILOSOPHY

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when philosophers searched for meaning in a troubled world. Henri Bergson wrote about a "vital impetus" that drives the evolution of life. Ludwig Wittgenstein came from a musical family, and his brother Paul, who lost an arm in the war, was a one-handed pianist. In 1920 Jan Łukasiewicz created a logic in which a statement can be true, false, or possible – a third value. He belonged to the Lwów–Warsaw School, a famous circle of Polish logicians and philosophers. In 1923, in "I and Thou," Martin Buber wrote that a person becomes truly themselves through encounter with another. Mahatma Gandhi fought for India's freedom through peaceful, nonviolent resistance. Edith Stein, a student of Husserl, died at Auschwitz in 1942; today she is a saint and a patron saint of Europe. Jean-Paul Sartre claimed that man is condemned to be free, and Viktor Frankl, who survived the camps, wrote that what a person needs most is meaning in life. Remember these ideas – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Philosopher Ludwig Wittgenstein came from a family visited by Brahms and Mahler, and his brother Paul was...**
 - a) a violinist
 - b) a one-handed pianist
 - c) a singer
 - d) a conductor
2. 👁 **The Łwów–Warsaw School was a famous circle of Polish...**
 - a) painters
 - b) soccer players
 - c) logicians and philosophers
 - d) architects
3. 🗨 **Existentialist Jean-Paul Sartre claimed that man is "condemned to be..."**
 - a) alone
 - b) happy
 - c) free
 - d) successful
4. ➡ **Jan Łukasiewicz created a logic in which a statement can be true, false, or...**
 - a) green
 - b) loud
 - c) possible (a third value)
 - d) forbidden
5. ✚ **Mahatma Gandhi fought for India's freedom through...**
 - a) peaceful, nonviolent resistance
 - b) assassinations
 - c) armed uprising
 - d) guerrilla warfare
6. 🗣 **In his book "I and Thou" (1923), Martin Buber wrote that a person becomes truly themselves through...**
 - a) encounter with another
 - b) war
 - c) solitude
 - d) work
7. ❤ **Viktor Frankl, who survived concentration camps, wrote that what a person needs most is...**
 - a) fame
 - b) meaning in life
 - c) power
 - d) wealth
8. 🌿 **Henri Bergson wrote about a "vital impetus" (élan vital) which, in his view, drives...**
 - a) politics
 - b) the evolution of life
 - c) steam engines
 - d) the stars
9. ☆ **Edith Stein, a philosopher and student of Husserl, died at Auschwitz in 1942. Today she is a saint and also...**
 - a) the patron saint of cooks
 - b) the patron saint of pilots
 - c) the patron saint of sailors
 - d) a patron saint of Europe

HISTORY

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars. In 1914, after the assassination in Sarajevo of Archduke Franz Ferdinand, World War I broke out. Soldiers of Józef Piłsudski's Legions sang "The First Brigade." On November 11, 1918, Poland regained its independence after the partitions, which had lasted from 1795 to 1918 – 123 years. After the war the League of Nations was founded to prevent wars. In August 1920 the Battle of Warsaw, known as the "Miracle on the Vistula," stopped the march of the Red Army. During World War II, Cavalry Captain Witold Pilecki voluntarily got himself sent to Auschwitz to gather information about the camp. The winter of 1941/42 helped the Red Army stop the Germans outside Moscow. The war ended in Europe on May 8, 1945, and worldwide in September 1945, after the surrender of Japan. Remember these events – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏳️ The song "The First Brigade" was sung by soldiers of...
 - a) the January Uprising
 - b) Napoleon's army
 - c) Jagiello's army
 - d) Józef Piłsudski's Legions
2. 👁️ World War I broke out after the assassination in Sarajevo (1914) of...
 - a) Kaiser Wilhelm II
 - b) the King of England
 - c) Archduke Franz Ferdinand
 - d) Tsar Nicholas II
3. 🗞️ Poland regained its independence on...
 - a) September 1, 1939
 - b) May 3, 1791
 - c) November 11, 1918
 - d) August 15, 1920
4. 🇵🇱 The partitions of Poland lasted from 1795 to 1918. How many years is that?

5. ⚔️ The Battle of Warsaw in August 1920 is known as the...
 - a) "Battle of the Nations"
 - b) "Blitzkrieg"
 - c) "Miracle on the Vistula"
 - d) "Night of Broken Glass"
6. 🌐 The League of Nations, the predecessor of the UN, was founded after World War I in order to...
 - a) build a railroad
 - b) print money
 - c) prevent wars
 - d) organize the Olympics
7. 💖 Cavalry Captain Witold Pilecki voluntarily got himself sent to Auschwitz in order to...
 - a) escape to Switzerland
 - b) gather information about the camp
 - c) find his family
 - d) work as a doctor
8. 🍁 The winter of 1941/42 helped the Red Army stop the Germans outside...
 - a) Paris
 - b) Moscow
 - c) Rome
 - d) London
9. ⭐ World War II ended in Europe on May 8, 1945, and worldwide in September 1945 after the surrender of...
 - a) Sweden
 - b) Italy
 - c) Japan
 - d) Spain

LINGUISTICS

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when words and codes decided people's fate. In World War I animals also carried messages, like the famous pigeon Cher Ami. The German Enigma cipher machine had a keyboard and lamps with letters, and it was hard to break because it changed the cipher after every letter. Even so, in 1932 the Polish mathematicians Rejewski, Różycki and Zygalski broke its cipher, and machines called "cryptologic bombs" helped break Enigma ciphers. Navajo men served in the US Army as code talkers using their own language. Esperanto was popular then, but in the Third Reich and the USSR Esperantists were persecuted. Julian Tuwim played with language in poems for children, for example in "The Locomotive" from 1938. And in 1943 "The Little Prince" reminded us that one sees clearly only with the heart. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The Enigma cipher machine had a keyboard and...
 - a) a loudspeaker
 - b) musical notes
 - c) a screen
 - d) lamps with letters
2. 👁 The Enigma cipher was broken in 1932 by Polish mathematicians Rejewski, Różycki and...
 - a) Curie
 - b) Banach
 - c) Zygański
 - d) Copernicus
3. 🗨 Navajo men served in the US Army as...
 - a) code talkers using their own language
 - b) pilots
 - c) Japanese interpreters
 - d) cooks
4. ➡ Enigma was hard to break because it...
 - a) used Roman numerals
 - b) was invisible
 - c) wrote backwards
 - d) changed the cipher after every letter
5. 💣 Machines called "cryptologic bombs" helped...
 - a) drop bombs
 - b) count votes
 - c) build bridges
 - d) break Enigma ciphers
6. 🗣 Esperanto was popular between the wars, but in the Third Reich and the USSR Esperantists were...
 - a) rewarded
 - b) elected to government
 - c) persecuted
 - d) hired by the radio
7. 💖 Julian Tuwim played with language in his poems for children, for example in the poem...
 - a) "Rota"
 - b) "The Locomotive"
 - c) "Pan Tadeusz"
 - d) "Ode to Joy"
8. 🐦 In World War I, animals also carried messages, for example the famous pigeon...
 - a) Pegasus
 - b) Cher Ami
 - c) Icarus
 - d) Robin Hood
9. ☆ Antoine de Saint-Exupéry's "The Little Prince" was published in 1943 and says that "one sees clearly only with..."
 - a) the eyes
 - b) a view from above
 - c) the heart
 - d) the night

LITERARY STUDIES

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars and open its books. In 1915, in "The Metamorphosis," Franz Kafka described a man who wakes up as a giant insect. In 1924 Władysław Reymont won the Nobel Prize for the novel "The Peasants." In 1926 A.A. Milne described the adventures of a bear named Winnie-the-Pooh. Kornel Makuszyński wrote about Koziółek Matolek, the goat who was searching for Pacanów. In 1929, in "All Quiet on the Western Front," Remarque told of young soldiers in World War I. In 1937 J.R.R. Tolkien published a novel about the hobbit Bilbo Baggins, and Witold Gombrowicz wrote "Ferdynand." During the war Anne Frank kept her diary – the journal of a girl who hid from the Nazis with her family. And in 1945, in "Animal Farm," George Orwell portrayed dictatorship disguised as a fable about animals. Remember these books – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏆 **Władysław Reymont won the Nobel Prize in 1924 for the novel...**
 - a) "Quo Vadis"
 - b) "The Spring to Come"
 - c) "The Doll"
 - d) "The Peasants"
2. 👁️ **In 1937, J.R.R. Tolkien published a novel about...**
 - a) the hobbit Bilbo Baggins
 - b) the wizard Harry
 - c) Aslan the lion
 - d) Winnie-the-Pooh
3. 🗨️ **The author of "Ferdynand" (1937) is...**
 - a) Czesław Miłosz
 - b) Witold Gombrowicz
 - c) Bruno Schulz
 - d) Stanisław Lem
4. 📖 **In "Animal Farm" (1945), George Orwell portrayed dictatorship disguised as...**
 - a) a detective story
 - b) a romance
 - c) a fable about animals
 - d) a space novel
5. 🐐 **Kornel Makuszyński wrote about Koziołek Matolek (Billy the Goat), who was searching for...**
 - a) Hollywood
 - b) Paris
 - c) Kraków
 - d) Pacanów
6. 🧑 **Remarque's "All Quiet on the Western Front" (1929) is about...**
 - a) a journey to America
 - b) life in Paris
 - c) young soldiers in World War I
 - d) an expedition to the Pole
7. 💖 **"The Diary of Anne Frank" is the journal of a girl who...**
 - a) was a pilot
 - b) was a queen
 - c) fled to America
 - d) hid from the Nazis with her family
8. 🐻 **In 1926, A.A. Milne described the adventures of a bear named...**
 - a) Colargol
 - b) Paddington
 - c) Winnie-the-Pooh
 - d) Uszatek
9. ⭐ **In "The Metamorphosis" (1915), Franz Kafka describes a man who wakes up as...**
 - a) a giant insect
 - b) a cat
 - c) a tree
 - d) a bird

CULTURE AND RELIGION

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, full of music, film and faith. In 1917 the apparitions at Fátima were received by three shepherd children. Early in the century jazz was born in New Orleans, and in the 1920s the Charleston dance was hugely popular. In 1927 the first talking picture, "The Jazz Singer," reached movie theaters. In 1928 Walt Disney created the character of Mickey Mouse, and in 1937 he released the first feature-length animated film, "Snow White." That same year Pablo Picasso painted "Guernica," protesting the bombing of that Spanish town. St. Faustina Kowalska recorded her revelations in her "Diary." St. Maximilian Kolbe founded the friary of Niepokalanów near Warsaw, and in 1941 at Auschwitz he gave his life for a fellow prisoner. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎷 Jazz was born in the early 20th century in...
 - a) Paris
 - b) London
 - c) New Orleans
 - d) Vienna
2. 👁 In 1937, Pablo Picasso painted a work protesting the bombing of the Spanish town of...
 - a) Seville
 - b) Guernica
 - c) Barcelona
 - d) Toledo
3. 🗨 Disney's first feature-length animated film (1937) was...
 - a) "Snow White"
 - b) "The Lion King"
 - c) "Cinderella"
 - d) "Pinocchio"
4. 🎬 The first talking picture ("The Jazz Singer") reached movie theaters in...
 - a) 1900
 - b) 1927
 - c) 1945
 - d) 1950
5. 💃 The Charleston dance was hugely popular in the...
 - a) 1990s
 - b) 1920s
 - c) 1860s
 - d) 2010s
6. 👤 The apparitions at Fátima (1917) were received by three...
 - a) soldiers
 - b) priests
 - c) teachers
 - d) shepherd children
7. 💖 St. Faustina Kowalska recorded her revelations in her...
 - a) "Chronicles"
 - b) "Letters"
 - c) "Diary"
 - d) "Confessions"
8. 🐰 In 1928, Walt Disney created the character of...
 - a) Bugs Bunny
 - b) Mickey Mouse
 - c) Donald Duck
 - d) Garfield the Cat
9. ⭐ In which camp did St. Maximilian Maria Kolbe give his life for a fellow prisoner in 1941?
 - a) Auschwitz
 - b) Dachau
 - c) Treblinka
 - d) Stutthof

ECONOMICS AND FINANCE

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when the economy saw booms and busts. In 1913 Henry Ford introduced the assembly line, which made cars cheaper. In 1923 hyperinflation in Germany meant that people burned banknotes in their stoves. In 1924 the reform of Władysław Grabski replaced the Polish mark with the złoty. Between the wars Gdynia was built as Poland's seaport, and Eugeniusz Kwiatkowski, creator of the Central Industrial Region, argued that Poland needed its own industry. In 1929 the New York stock market crash began the Great Depression, and the songs of those years talked about poverty and hope for better times. In the 1930s droughts and dust storms in the US caused farmers to flee the Great Plains. After the war, from 1948, the Marshall Plan brought American aid for rebuilding Europe after the war. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏳️ **Songs from the Great Depression often talked about...**
 - a) poverty and hope for better times
 - b) sports cars
 - c) vacations
 - d) wealth
2. 👁️ **The Great Depression began with the New York stock market crash in...**
 - a) 1939
 - b) 1929
 - c) 1945
 - d) 1914
3. 🗨️ **The Polish złoty replaced the Polish mark in 1924 thanks to the reform of...**
 - a) Józef Piłsudski
 - b) Roman Dmowski
 - c) Ignacy Paderewski
 - d) Władysław Grabski
4. 📈 **In Germany in 1923, hyperinflation meant that...**
 - a) people burned banknotes in their stoves
 - b) there was no money at all
 - c) prices were falling
 - d) money was made of gold
5. ⚙️ **In 1913, Henry Ford introduced the assembly line, which made...**
 - a) working hours longer
 - b) factories close
 - c) cars cheaper
 - d) cars able to fly
6. 🏰 **Between the wars, Gdynia was built as...**
 - a) the capital of Poland
 - b) a fortress
 - c) a mountain resort
 - d) Poland's seaport
7. 💚 **Eugeniusz Kwiatkowski, creator of the Central Industrial Region (COP), argued that Poland needed...**
 - a) more horses
 - b) to close its ports
 - c) its own industry
 - d) fewer schools
8. 🌾 **Droughts and dust storms in the US in the 1930s (the Dust Bowl) caused...**
 - a) farmers to flee the Great Plains
 - b) new forests
 - c) higher crop yields
 - d) floods
9. ⭐ **The Marshall Plan (from 1948) was American aid for...**
 - a) building rockets
 - b) conquering Mars
 - c) rebuilding Europe after the war
 - d) the Olympic Games

PSYCHOLOGY

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when psychology explored the human mind and behavior. Ivan Pavlov showed that dogs can salivate at the sound of a bell, if they associate it with food. Carl Gustav Jung introduced the concepts of the introvert and the extrovert. In 1921 the Rorschach test appeared, based on interpreting inkblots. Jean Piaget studied how children develop their thinking, and Janusz Korczak believed that a child is a full human being. "Shell shock" was the psychological suffering of soldiers after their experiences at the front, and music therapy developed when people noticed that music helped wounded soldiers in hospitals. In 1943 Abraham Maslow described the hierarchy of needs pyramid. Konrad Lorenz discovered that goslings accept as their mother the first creature they see. And after the war psychologists studied why ordinary people carried out cruel orders. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎵 **Music therapy began to develop when people noticed that music helped...**
 - a) water plants
 - b) wounded soldiers in hospitals
 - c) birds in flight
 - d) machines at work
2. 👁️ **The Rorschach test (1921) involves interpreting...**
 - a) photographs
 - b) inkblots
 - c) dreams
 - d) poems
3. 🗨️ **Carl Gustav Jung introduced the concepts of the introvert and the...**
 - a) extrovert
 - b) genius
 - c) hero
 - d) choleric
4. 🐶 **In his experiments, Ivan Pavlov showed that dogs can salivate at the sound of...**
 - a) a cat
 - b) a bell, if they associate it with food
 - c) their own name, always
 - d) rain
5. 👶 **Jean Piaget studied how children...**
 - a) sleep
 - b) eat
 - c) grow physically
 - d) develop their thinking
6. 👤 **Janusz Korczak believed that a child...**
 - a) has no rights
 - b) is a full human being
 - c) is a small adult without feelings
 - d) must only obey
7. 💖 **Who created the "hierarchy of needs" pyramid?**
 - a) Sigmund Freud
 - b) Ivan Pavlov
 - c) Carl Gustav Jung
 - d) Abraham Maslow
8. 🐣 **Konrad Lorenz discovered that goslings accept as their mother...**
 - a) the loudest sound
 - b) the first creature they see
 - c) the biggest animal
 - d) water
9. ⭐ **After World War II, psychologists studied why ordinary people...**
 - a) liked sports
 - b) read books
 - c) rode bicycles
 - d) carried out cruel orders

EDUCATION

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when schools were searching for new paths. Maria Montessori's method is based on the child's own activity in a prepared setting. Forest schools held their lessons in the open air. Carl Orff taught children music using simple instruments, rhythm and movement. Around 1910 Polish scouting was born, modeled on the scouting of Robert Baden-Powell, who taught that the goal is to leave the world a little better. In 1919 Poland introduced compulsory schooling for children aged 7 to 14. Janusz Korczak founded in Warsaw an Orphans' Home with child self-government, and in 1942 he stayed with the children in his care to the very end. During the occupation there were secret classes – underground lessons held in homes – and the Grey Ranks carried out acts of minor sabotage. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎵 **Carl Orff created a method of teaching music to children using...**
 - a) simple instruments, rhythm and movement
 - b) lectures
 - c) only the piano
 - d) reading sheet music
2. 👁 **Polish scouting (harcerstwo) began around 1910 and was modeled on the scouting movement of...**
 - a) Józef Piłsudski
 - b) Robert Baden-Powell
 - c) Maria Montessori
 - d) Janusz Korczak
3. 🗨 **In Warsaw, Janusz Korczak founded for orphans...**
 - a) an orphanage without rules
 - b) a military school
 - c) a university
 - d) an Orphans' Home with child self-government
4. 🇵🇱 **In 1919, Poland introduced...**
 - a) free education for boys only
 - b) a total ban on village schools
 - c) traveling schools on horseback
 - d) compulsory schooling for children aged 7 to 14
5. ⚔ **The Grey Ranks were underground scouts who during the war carried out, among other things...**
 - a) regattas
 - b) acts of minor sabotage
 - c) expeditions to the Himalayas
 - d) dance schools
6. 👤 **Maria Montessori's educational method is based above all on...**
 - a) the child's own activity in a prepared setting
 - b) memorizing and repeating texts by heart
 - c) learning only in large groups with lectures
 - d) a strict system of punishments and rewards
7. ❤ **In 1942, Janusz Korczak...**
 - a) stayed with the children in his care
 - b) left for Palestine
 - c) fled abroad
 - d) became minister of education
8. 🌿 **Forest schools in the early 20th century held their lessons...**
 - a) in the open air
 - b) in basements
 - c) at night
 - d) on ships
9. ☆ **According to Baden-Powell, the goal of scouting was to...**
 - a) leave the world a little better
 - b) earn as much money as possible
 - c) win wars against neighbors
 - d) gain power over others

SOCIOLOGY

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when societies changed at a rapid pace. In Britain the suffragettes fought for women's voting rights. During World War I women on a massive scale replaced men in factories, and in 1918 women in Poland gained the right to vote. From 1918 to 1920 the Spanish flu killed tens of millions of people worldwide. In the US came the Great Migration of Black Americans from the South to cities in the North. Radio became a window on the world for millions of families. In 1940, during World War II, the Germans created the Warsaw Ghetto, and in April 1943 an uprising broke out there. Irena Sendler and her helpers smuggled about 2,500 children out of the ghetto. And the Ulma family from Markowa were killed in 1944 because they were hiding Jews. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎵 **Between the wars, radio became...**
 - a) banned in most European countries
 - b) a forgotten invention of the Middle Ages
 - c) a window on the world for millions of families
 - d) a luxury reserved only for kings
2. 👁 **Women who fought for women's voting rights in Britain were called...**
 - a) musketeers
 - b) suffragettes
 - c) ladies-in-waiting
 - d) feminists
3. 🗨 **The Great Migration in the US was the move of millions of Black Americans from the South to...**
 - a) Africa
 - b) Mexico
 - c) Canada
 - d) cities in the North
4. 🗳 **Women in Poland gained the right to vote in...**
 - a) 1918
 - b) 1989
 - c) 1945
 - d) 1791
5. ⚔ **During World War I, women on a massive scale...**
 - a) stopped working
 - b) closed the schools
 - c) replaced men in factories
 - d) left the cities
6. 🧑 **Irena Sendler and her helpers smuggled out of the Warsaw Ghetto about...**
 - a) 25 children
 - b) 2,500 children
 - c) 250,000 children
 - d) 5 children
7. ❤ **The Ulma family from Markowa were killed in 1944 because they...**
 - a) stole horses
 - b) did not pay taxes
 - c) were hiding Jews
 - d) deserted from the army
8. 🌿 **The Spanish flu epidemic (1918–1920) killed worldwide...**
 - a) only animals
 - b) a thousand people, only in Spain
 - c) nobody
 - d) tens of millions of people
9. ⭐ **The Warsaw Ghetto was created in 1940. In April 1943, what broke out there was...**
 - a) a wedding
 - b) a festival
 - c) the Olympic Games
 - d) an uprising

LAW

**8. Era of the World Wars**

1900–1950 AD

Let's travel back to the era of the two world wars, when law tried to bring order to the world. In 1919, at the palace of Versailles near Paris, the treaty ending World War I with Germany was signed. The March Constitution of 1921 made Poland a parliamentary republic. In 1927 "Dąbrowski's Mazurka" officially became Poland's national anthem. The Geneva Convention of 1929 obliged states to treat prisoners of war humanely. In 1932 Pieniny National Park was created, one of the first Polish parks established to protect nature. In 1945 the United Nations was founded, and Poland's borders were shifted to the west. In 1945–1946 the Nuremberg trials tried the leaders of Nazi Germany. The Polish lawyer Raphael Lemkin coined the word genocide, and in 1948 the UN General Assembly adopted the Universal Declaration of Human Rights. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 "Dąbrowski's Mazurka" officially became Poland's national anthem in...
a) 1989
b) 1797
c) 1927
d) 1918
2. 👁️ The treaty ending World War I with Germany was signed in 1919 at the palace of...
a) Buckingham in London
b) Schönbrunn in Vienna
c) Versailles near Paris
d) the Kremlin in Moscow
3. 📜 The March Constitution of 1921 made Poland...
a) a colony
b) a monarchy
c) an empire
d) a parliamentary republic
4. 🇺🇳 The United Nations was founded in...
a) 1939
b) 1989
c) 1919
d) 1945
5. ⚖️ The Nuremberg trials (1945–1946) tried...
a) kings of England
b) the leaders of Nazi Germany
c) inventors
d) soccer players
6. 🌍 Who adopted the Universal Declaration of Human Rights in 1948?
a) The Council of Europe
b) The European Parliament
c) The League of Nations
d) The UN General Assembly
7. ❤️ The Geneva Convention of 1929 obliged states to treat humanely...
a) athletes
b) writers
c) prisoners of war
d) artists
8. 🌿 The first Polish national parks, such as Pieniny (1932), were created to protect...
a) factories
b) mines
c) nature
d) castles
9. ⭐ After World War II ended, Poland's borders were...
a) left unchanged
b) shifted to the west
c) erased
d) shifted to the east

OTHER (HUMANITIES)



8. Era of the World Wars

1900–1950 AD

Let's travel back to the era of the two world wars, and more precisely to the years of World War II, when Poles fought on many fronts. The symbol of the underground was the Fighting Poland anchor, made up of the letters P and W. The Pole Józef Kosacki invented a mine detector that saved many soldiers. In 1944 General Anders's soldiers fought at Monte Cassino in Italy. Wojtek the bear carried ammunition crates there, and the song "The Red Poppies on Monte Cassino" was written the night before the monastery was captured. On August 1, 1944, at "W" Hour, which was 5:00 p.m., the Warsaw Uprising broke out. It lasted until October 2 – 63 days. The pianist Władysław Szpilman survived the war with help from, among others, the German officer Wilm Hosenfeld. And people who saved Jews at the risk of their own lives receive from the Yad Vashem Institute the title of Righteous Among the Nations. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏳️ **Władysław Szpilman, the pianist whose story is told in the film "The Pianist," survived the war with help from, among others...**
 - a) the American army
 - b) German officer Wilm Hosenfeld
 - c) the King of England
 - d) the Pope
2. 👁️ **The Fighting Poland anchor is a symbol made up of the letters...**
 - a) R and P
 - b) A and K
 - c) S and G
 - d) P and W
3. 🗨️ **The Warsaw Uprising broke out on August 1, 1944, at "W" Hour, which was...**
 - a) 12:00 noon
 - b) 9:00 p.m.
 - c) 6:00 a.m.
 - d) 5:00 p.m.
4. 📅 **The Warsaw Uprising lasted from August 1 to October 2, 1944. How many days is that?**

5. 🐻 **At Monte Cassino, Wojtek the bear from General Anders's army...**
 - a) fought with a bayonet
 - b) cooked soup
 - c) flew a plane
 - d) carried ammunition crates
6. 🌍 **The Battle of Monte Cassino (1944), in which Poles fought, took place in...**
 - a) Italy
 - b) France
 - c) Norway
 - d) Africa
7. 💖 **The song "The Red Poppies on Monte Cassino" was written...**
 - a) after the war in London
 - b) the night before the monastery was captured
 - c) in the 19th century
 - d) in Warsaw in 1920
8. 🌿 **During the war, Poles invented a mine detector. Its inventor was...**
 - a) Stefan Kudelski
 - b) Ignacy Łukasiewicz
 - c) Jan Szczepanik
 - d) Józef Kosacki
9. ⭐ **People who saved Jews at the risk of their own lives receive from the Yad Vashem Institute the title of...**
 - a) Hero of Labor
 - b) Righteous Among the Nations
 - c) Honorary Citizen
 - d) Knight of the Round Table

ARCHITECTURE AND URBAN PLANNING

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when architecture was searching for new forms. In 1913 the Centennial Hall in Wrocław had the world's largest dome made of reinforced concrete. In 1919 in Weimar Walter Gropius founded the Bauhaus design school. Le Corbusier called a house "a machine for living in." In the 1920s and 1930s the Art Deco style was fashionable, decorated with geometric patterns and shine, and Gdynia became famous for its modernist architecture. In 1931 the Empire State Building was completed after only about 13 months of construction. In 1935 Frank Lloyd Wright designed the house "Fallingwater," which stands over a waterfall. After the war the concert hall of the Warsaw Philharmonic had to be rebuilt from ruins, and Warsaw's Old Town was rebuilt using, among other things, paintings by Canaletto. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏏 **After the war, the concert hall of the Warsaw Philharmonic (opened in 1901) had to be...**
 - a) rebuilt from ruins
 - b) turned into a train station
 - c) moved to Kraków
 - d) demolished forever
2. 👁 **The Art Deco style, fashionable in the 1920s and 1930s, is decorated with...**
 - a) thatched roofs
 - b) geometric patterns and shine
 - c) Gothic arches
 - d) Baroque angels
3. 🗨 **The Bauhaus design school was founded in Weimar in 1919 by...**
 - a) Frank Lloyd Wright
 - b) Walter Gropius
 - c) Antoni Gaudí
 - d) Le Corbusier
4. 🏠 **Le Corbusier called a house...**
 - a) "a palace of dreams"
 - b) "a temple of the family"
 - c) "a machine for living in"
 - d) "a cave of the spirit"
5. ⚡ **The construction of the Empire State Building (1931) took only about...**
 - a) 13 years
 - b) 13 days
 - c) 13 months
 - d) 130 years
6. 🧑 **Interwar Gdynia is famous for its...**
 - a) Romanesque architecture
 - b) Gothic architecture
 - c) modernist architecture
 - d) Baroque architecture
7. 💖 **When it was built, the Centennial Hall in Wrocław (1913) had the world's largest dome made of...**
 - a) wood
 - b) glass
 - c) stone
 - d) reinforced concrete
8. 🌿 **Frank Lloyd Wright designed the house "Fallingwater" (1935), which stands...**
 - a) over a waterfall
 - b) on a glacier
 - c) underground
 - d) on a skyscraper roof
9. ⭐ **After the war, Warsaw's Old Town was rebuilt using, among other things...**
 - a) computers
 - b) paintings by Canaletto
 - c) Roman plans
 - d) satellite images



8. Era of the World Wars

1900–1950 AD

Let's travel back to the era of the two world wars, when electronics was born. In 1906 the vacuum tube, the triode, made it possible to amplify sound in radios. Jan Szczepanik, the "Polish Edison," invented among other things a bulletproof vest made of silk. In 1920 Karel Čapek popularized the word "robot" in the play R.U.R., and Leon Theremin invented the theremin, played without touching it, by moving your hands in the air. In 1926 John Logie Baird showed a mechanical television, and in 1927 Philo Farnsworth an electronic one. Before the war Janusz Groszkowski worked mainly on radio engineering and vacuum tubes. Radar detects aircraft by sending out radio waves and receiving their reflection. At Bletchley Park Alan Turing designed the "Bombe" for breaking Enigma ciphers. In 1946 the ENIAC computer took up an entire large room, and in 1947 the transistor replaced vacuum tubes. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The vacuum tube (triode) of 1906 made it possible to...**
 - a) amplify sound in radios
 - b) freeze food
 - c) print books
 - d) build cars
2. 👁️ **Philo Farnsworth demonstrated the first electronic television in 1927, but a mechanical one had already been shown in 1926 by...**
 - a) Nikola Tesla
 - b) John Logie Baird
 - c) Thomas Edison
 - d) Guglielmo Marconi
3. 🗨️ **Who popularized the word "robot" in the play R.U.R. (1920)?**
 - a) Isaac Asimov
 - b) Stanisław Lem
 - c) Jules Verne
 - d) Karel Čapek
4. 📡 **Radar detects aircraft by sending out radio waves and receiving...**
 - a) their light
 - b) their smell
 - c) their reflection
 - d) the sound of their engines
5. 🔌 **The transistor was invented at Bell Labs in 1947. It replaced...**
 - a) engines
 - b) vacuum tubes
 - c) light bulbs
 - d) batteries
6. 🧑‍💻 **The first electronic computer, ENIAC (1946), took up...**
 - a) a pocket
 - b) a desk
 - c) an entire large room
 - d) a suitcase
7. 💖 **At Bletchley Park, Alan Turing designed a machine called the "Bombe" for...**
 - a) forecasting the weather
 - b) breaking Enigma ciphers
 - c) making ice cream
 - d) building roads
8. 🍀 **Before the war, Polish engineer Janusz Groszkowski worked mainly on...**
 - a) painting
 - b) horse breeding
 - c) radio engineering and vacuum tubes
 - d) poetry
9. ⭐ **Jan Szczepanik, the "Polish Edison," invented among other things...**
 - a) the telephone
 - b) dynamite
 - c) the light bulb
 - d) a bulletproof vest made of silk

COMPUTER SCIENCE AND TELECOMMUNICATIONS

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when radio waves connected the world. On Christmas Eve 1906 Reginald Fessenden transmitted the first radio broadcast with music and speech. In 1926 Polish Radio began regular broadcasting. In 1927 the transatlantic telephone connection worked thanks to radio waves. Marian Rejewski deciphered Enigma using the mathematical theory of permutations. In 1939 the Gleiwitz radio station was the site of a German provocation before the attack on Poland. During the war the resistance listened to the BBC, where hidden messages came as odd sentences, e.g. "the long sobs of the violins." In 1941 in Germany Konrad Zuse built the programmable computer Z3. Grace Hopper, working on the Mark II computer, popularized the word "bug," meaning an error. In 1948 Claude Shannon founded information theory, whose basic unit is the bit, and in 1950 Alan Turing proposed a test of whether a machine can converse like a human. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎧 In what year did Polish Radio begin regular broadcasting?
 - a) 1945
 - b) 1939
 - c) 1926
 - d) 1918
2. 🖥️ In 1941 in Germany, Konrad Zuse built the first working programmable computer, called the...
 - a) Z3
 - b) Enigma
 - c) Apple I
 - d) ENIAC
3. 🗨️ In 1950, Alan Turing proposed a test to check whether a machine...
 - a) can converse like a human
 - b) can calculate fast
 - c) can play chess
 - d) can fly
4. 📡 Claude Shannon founded information theory (1948), and the basic unit of information is the...
 - a) hertz
 - b) bit
 - c) liter
 - d) kilogram
5. 🎵 During the war, the resistance listened to the BBC, which passed on hidden messages in the form of...
 - a) hidden musical notes in symphonies
 - b) odd sentences, e.g. "the long sobs of the violins"
 - c) Morse code in pop songs
 - d) photographs described aloud
6. 🐼 Grace Hopper worked on the Mark II computer and popularized the word...
 - a) "internet"
 - b) "mouse"
 - c) "app"
 - d) "bug" (error)
7. 💖 Rejewski deciphered Enigma using the mathematical theory of...
 - a) permutations
 - b) fractions
 - c) geometry
 - d) dice probability
8. 🌿 The transatlantic telephone connection (1927) worked thanks to...
 - a) pigeons
 - b) satellites
 - c) radio waves
 - d) copper cables on the seabed
9. 🌟 In 1939, the Gleiwitz (Gliwice) radio station was the site of...
 - a) the Titanic's SOS call
 - b) a German provocation before the attack on Poland
 - c) a Chopin concert
 - d) the first children's program

BIOMEDICAL ENGINEERING



8. Era of the World Wars

1900–1950 AD

Let's travel back to the era of the two world wars, when technology began to save lives. In 1901 Karl Landsteiner discovered blood groups A, B and O, and the Polish physician Ludwik Hirszfeld introduced the labels A, B, AB and O. In 1903 Willem Einthoven built the electrocardiograph, the ECG. During World War I Marie Skłodowska-Curie organized mobile X-ray units. Prosthetic limbs were mass-produced because many soldiers had lost limbs, and the surgeon Harold Gillies developed facial plastic surgery. From 1928 the iron lung helped patients with polio to breathe. In 1931 the electron microscope made it possible to see viruses for the first time. And during World War II Willem Kolff built the first artificial kidney. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The electrocardiograph (ECG), which records the heart's electrical activity, was built in 1903 by...**
 - a) Wilhelm Röntgen
 - b) Willem Einthoven
 - c) Alexander Fleming
 - d) Louis Pasteur
2. 👁 **During World War I, Marie Skłodowska-Curie organized...**
 - a) a flight school
 - b) a hospital for horses
 - c) an aircraft factory
 - d) mobile X-ray units
3. 🗨 **The iron lung (1928) helped patients with...**
 - a) smallpox to see
 - b) polio to breathe
 - c) flu to eat
 - d) tuberculosis to walk
4. ➡ **Blood groups A, B and O were discovered in 1901 by...**
 - a) Paul Ehrlich
 - b) Robert Koch
 - c) Ludwik Hirsfeld
 - d) Karl Landsteiner
5. ⚡ **The first artificial kidney (dialysis machine) was built during the war by...**
 - a) Christiaan Barnard
 - b) Willem Kolff
 - c) Jonas Salk
 - d) Ludwik Hirsfeld
6. 🧑 **Polish physician Ludwik Hirsfeld introduced the blood group labels...**
 - a) 1, 2 and 3
 - b) X and Y
 - c) alpha and beta
 - d) A, B, AB and O
7. ❤ **Prosthetic limbs for World War I veterans were mass-produced because...**
 - a) they were in fashion
 - b) the League of Nations ordered it
 - c) many soldiers had lost limbs
 - d) there was too much wood
8. 🌿 **The electron microscope (1931) made it possible for the first time to see...**
 - a) planets
 - b) stars
 - c) viruses
 - d) bones
9. ⭐ **After World War I, surgeon Harold Gillies developed...**
 - a) laser surgery
 - b) facial plastic surgery
 - c) heart transplants
 - d) dentistry

CHEMICAL ENGINEERING

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when chemistry gave the world new materials. In 1907 Bakelite appeared, one of the first plastics, and gramophone records were made mainly of shellac. The Haber–Bosch process made it possible to produce ammonia for fertilizers from nitrogen in the air. Yet Fritz Haber also created chemical weapons, which shows that inventions can serve both good and evil. Poison gases such as mustard gas were first used on a mass scale in World War I. In 1916 Jan Czochralski developed a method for growing crystals. In 1935 nylon appeared and became famous for stockings. During World War II synthetic rubber was developed because natural rubber ran short, and from the 1940s the pesticide DDT helped fight mosquitoes that carry malaria. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Gramophone records of that period were made mainly of...**
 - a) metal
 - b) shellac
 - c) plastic vinyl
 - d) glass
2. 👁 **Nylon, the first fully synthetic fiber (1935), became famous for...**
 - a) umbrellas
 - b) hats
 - c) stockings
 - d) shoes
3. 🗨 **Bakelite (1907) was one of the first...**
 - a) metals
 - b) plastics
 - c) medicines
 - d) dyes
4. ⚖ **The Haber–Bosch process made it possible to produce, from nitrogen in the air...**
 - a) gold
 - b) rocket fuel
 - c) ammonia for fertilizers
 - d) oxygen for breathing
5. ⚡ **In 1916, Jan Czochochalski developed a method for growing...**
 - a) fish
 - b) bacteria
 - c) crystals
 - d) plants
6. 🧑 **Synthetic rubber was developed during World War II because...**
 - a) rubber went out of fashion
 - b) natural rubber ran short
 - c) doctors ordered it
 - d) it was cheaper than water
7. ❤ **Fritz Haber created fertilizers but also chemical weapons. This shows that inventions...**
 - a) do not matter
 - b) are always bad
 - c) can serve both good and evil
 - d) are always good
8. 🌿 **The pesticide DDT, used from the 1940s, initially helped fight...**
 - a) rain
 - b) birds
 - c) mosquitoes that carry malaria
 - d) fish
9. ⭐ **Poison gases such as mustard gas were first used on a mass scale in...**
 - a) the Napoleonic Wars
 - b) the 21st century
 - c) ancient times
 - d) World War I

CIVIL ENGINEERING AND TRANSPORT

**8. Era of the World Wars**

1900–1950 AD

Let's travel back to the era of the two world wars, when people took to the air. In 1903 the Wright brothers made the first powered airplane flight. In 1912, 9 years later, the Titanic struck an iceberg and sank. In 1914 the Panama Canal opened, connecting the Atlantic with the Pacific. In 1927 Charles Lindbergh became the first person to fly solo and nonstop across the Atlantic, and in 1932 Amelia Earhart became the first woman to fly solo across the Atlantic. That same year the Polish aviators Żwirko and Wigura won the Challenge air competition. In 1937 the airship Hindenburg, filled with hydrogen, burned. And in 1940 No. 303 Squadron, made up of Polish pilots, fought in the Battle of Britain. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The Wright brothers made the first powered airplane flight in...**
 - a) 1803
 - b) 1903
 - c) 1853
 - d) 1953
2. 👁 **The Panama Canal (1914) connects...**
 - a) the Baltic with the North Sea
 - b) the Atlantic with the Pacific
 - c) the Mediterranean with the Red Sea
 - d) the Danube with the Rhine
3. 🗨 **In 1927, Charles Lindbergh became the first person to fly solo and nonstop across...**
 - a) the Pole
 - b) the Atlantic
 - c) the Pacific
 - d) the Sahara
4. 🚢 **The Titanic sank in 1912, and the Wright brothers' plane took off in 1903. How many years apart were these events?**

5. 🚀 **In 1932, Polish aviators Żwirko and Wigura...**
 - a) built a rocket
 - b) flew to the Moon
 - c) flew over the Pole
 - d) won the Challenge air competition
6. 🇫🇷 **No. 303 Squadron in the Battle of Britain (1940) was made up of...**
 - a) French tank crews
 - b) Polish pilots
 - c) American sailors
 - d) British cavalrymen
7. 💖 **In 1932, Amelia Earhart became the first woman to...**
 - a) swim across the Atlantic
 - b) build an airplane
 - c) fly into space
 - d) fly solo across the Atlantic
8. 🍃 **The airship Hindenburg burned in 1937 because it was filled with...**
 - a) air
 - b) helium
 - c) hydrogen
 - d) steam
9. ☆ **In 1912, the Titanic struck...**
 - a) an iceberg
 - b) a rock
 - c) a whale
 - d) another ship

MATERIALS ENGINEERING



8. Era of the World Wars

1900–1950 AD

Let's travel back to the era of the two world wars, when new materials changed technology – and war. Around 1913 stainless steel was developed in Sheffield, England. Duralumin, an aluminum alloy lighter than steel, was important for building airplanes and airships. From 1916 tanks crossed mud and trenches thanks to their tracks. Eugène Freyssinet's prestressed concrete made it possible to build longer and thinner bridges. In 1938 Teflon was discovered by accident; today it is known for nonstick frying pans. During the war Plexiglas was used for aircraft cockpits, and tires were made from synthetic rubber when natural rubber ran out. Uranium, used in the atomic bomb, is a radioactive element, and in August 1945 the first atomic bombs were dropped on Hiroshima and Nagasaki. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Stainless steel was developed around 1913 in...**
 - a) Rome
 - b) Warsaw
 - c) Tokyo
 - d) Sheffield, England
2. 👁 **During the war, Plexiglas (acrylic glass) was used for...**
 - a) church roofs
 - b) aircraft cockpits
 - c) coins
 - d) shoe soles
3. 🗨 **Duralumin, an aluminum alloy lighter than steel, was important for building...**
 - a) books
 - b) clay huts
 - c) airplanes and airships
 - d) wooden boats
4. 🚧 **Tanks can cross mud and trenches thanks to their...**
 - a) skids
 - b) tracks
 - c) four wheels
 - d) wings
5. ⚡ **Prestressed concrete, invented by Eugène Freyssinet, made it possible to build...**
 - a) lighter airplanes
 - b) submarines
 - c) longer and thinner bridges
 - d) taller trees
6. 🧑 **Teflon was discovered by accident in 1938. Today it is known for...**
 - a) tires
 - b) glues
 - c) winter clothing
 - d) nonstick frying pans
7. ❤ **Synthetic rubber made it possible to produce tires when...**
 - a) horses came back
 - b) natural rubber ran out
 - c) cars had wings
 - d) tires went out of fashion
8. 🍃 **Uranium, used in the atomic bomb, is an element that is...**
 - a) noble like gold
 - b) liquid like mercury
 - c) as light as helium
 - d) radioactive
9. ⭐ **The first atomic bombs were dropped in August 1945 on...**
 - a) Berlin and Rome
 - b) Hiroshima and Nagasaki
 - c) London and Paris
 - d) Moscow and Kyiv

MECHANICAL ENGINEERING



8. Era of the World Wars

1900–1950 AD

Let's travel back to the era of the two world wars, when machines picked up speed. A car's combustion engine needs a muffler to reduce exhaust noise. In 1926 Robert Goddard launched the first rocket powered by liquid fuel. In the 1930s the Polski Fiat 508 was made in Warsaw under an Italian license, and from 1935 the Polish 7TP tank with a diesel engine. Electric refrigerators meant that food stayed fresh longer. In 1939 Igor Sikorsky built a successful helicopter, which stays in the air thanks to a spinning rotor. The jet engine was developed independently by Frank Whittle and Hans von Ohain. In 1944 the V-2 rocket became the first missile that reached space. Many inventions, such as radar and jet planes, developed quickly because the military needed them. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 A car's combustion engine needs a muffler in order to...
 - a) give off light
 - b) reduce exhaust noise
 - c) play tunes
 - d) increase speed
2. 👁️ A 1930s Polish car made in Warsaw under an Italian license was the...
 - a) Maluch
 - b) Polski Fiat 508
 - c) Polonez
 - d) Syrena
3. 🗨️ The jet engine was developed independently by Frank Whittle and...
 - a) Rudolf Diesel
 - b) Hans von Ohain
 - c) Karl Benz
 - d) Henry Ford
4. 🚁 A helicopter stays in the air thanks to...
 - a) bird-like wings
 - b) a balloon
 - c) a spinning rotor
 - d) a rocket
5. 🚀 The V-2 rocket was the first missile that...
 - a) flew underwater
 - b) reached space
 - c) ran on rails
 - d) floated
6. 🛡️ The Polish 7TP tank, produced from 1935, had...
 - a) wings
 - b) steam power
 - c) bicycle wheels
 - d) a diesel engine
7. ❤️ In 1926, Robert Goddard launched the first rocket powered by...
 - a) coal
 - b) gunpowder
 - c) steam
 - d) liquid fuel
8. 🌿 Electric refrigerators became common in the 1930s, which meant that...
 - a) stores disappeared
 - b) food stayed fresh longer
 - c) ice was banned
 - d) bread prices rose
9. ⭐ Many inventions, such as radar and jet planes, developed quickly because...
 - a) they were cheap
 - b) they were banned
 - c) they were simple
 - d) the military needed them



8. Era of the World Wars

1900–1950 AD

Let's travel back to the era of the two world wars, when the world was hungry for energy. Back in 1905 Einstein wrote the formula $E = mc^2$, which says that mass is a form of energy. Oil powered cars, ships and planes, and miners in Silesia mainly extracted hard coal. Poland's President Ignacy Mościcki was also an inventor and chemist. In 1930 the deadly smog in Belgium's Meuse Valley was caused by factory smoke in windless weather. In 1936 the Hoover Dam was completed; it produces electricity for cities. In the 1930s rural electrification in the US meant that farmers had light and machines. In 1942 in Chicago the team of Enrico Fermi built the first nuclear reactor. And the Manhattan Project was an American program to build the atomic bomb. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The Hoover Dam (1936) produces...**
 - a) electricity for cities
 - b) salt water
 - c) waterfall sounds
 - d) oil
2. 👁️ **The first nuclear reactor (Chicago, 1942) was built by the team of...**
 - a) Enrico Fermi
 - b) Albert Einstein
 - c) Marie Curie
 - d) Nikola Tesla
3. 🗨️ **Einstein's formula $E = mc^2$ says that mass is...**
 - a) the force of wind
 - b) an amount of water
 - c) a kind of light
 - d) a form of energy
4. ⚙️ **Oil became very important in the 20th century because it powered...**
 - a) cars, ships and planes
 - b) horses
 - c) windmills
 - d) sailboats
5. ⚔️ **Between the wars, miners in Silesia mainly extracted...**
 - a) hard coal
 - b) salt
 - c) diamonds
 - d) gold
6. 🧑🏫 **Rural electrification in many countries, e.g. the US in the 1930s, meant that...**
 - a) candle prices rose
 - b) people moved away
 - c) villages vanished
 - d) farmers had light and machines
7. 💖 **The deadly smog in Belgium's Meuse Valley (1930) was caused by...**
 - a) an earthquake
 - b) a volcanic eruption
 - c) factory smoke in windless weather
 - d) a forest fire
8. 🍀 **Ignacy Mościcki, President of Poland, was also...**
 - a) an actor
 - b) a pilot
 - c) a soccer player
 - d) an inventor and chemist
9. ⭐ **The Manhattan Project was an American program to build...**
 - a) a subway network
 - b) a skyscraper
 - c) the atomic bomb
 - d) a canal

ASTRONOMY



8. Era of the World Wars

1900–1950 AD

Let's look at the sky in the era of the two world wars. In 1908 the Tunguska event struck Siberia – probably a meteoroid exploding in the air. In 1915, in his general theory of relativity, Einstein predicted that starlight bends near the Sun. In 1916 Gustav Holst composed the suite "The Planets." In 1925 Cecilia Payne discovered that stars are made mainly of hydrogen and helium. In the 1920s Edwin Hubble proved that there are other galaxies beyond the Milky Way, and in 1927 Georges Lemaître proposed that the Universe is expanding from an initial state. In 1930 Clyde Tombaugh discovered Pluto. The Polish astronomer Tadeusz Banachiewicz studied the Moon's motion and eclipses. And in 1938 Orson Welles's radio drama "The War of the Worlds" caused confusion among some listeners. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In 1916, Gustav Holst composed the suite...
 - a) "Boléro"
 - b) "The Planets"
 - c) "The Four Seasons"
 - d) "The Nutcracker"
2. 👁 In the 1920s, Edwin Hubble proved that...
 - a) the Earth is flat
 - b) the Sun orbits the Earth
 - c) stars do not exist
 - d) there are other galaxies beyond the Milky Way
3. 📖 The planet Pluto was discovered in 1930 by...
 - a) Albert Einstein
 - b) Edwin Hubble
 - c) Clyde Tombaugh
 - d) Galileo
4. ⚡ In his general theory of relativity (1915), Einstein predicted that starlight...
 - a) does not exist
 - b) bends near the Sun
 - c) is green
 - d) goes out at night
5. ⚡ In 1927, Georges Lemaître proposed that the Universe...
 - a) is flat
 - b) is eternal and unchanging
 - c) is expanding from an initial state
 - d) spins in circles
6. 🧑 The Tunguska event (1908) in Siberia was probably...
 - a) a meteoroid exploding in the air
 - b) a bomb explosion
 - c) a volcanic eruption
 - d) an alien attack
7. ❤ Tadeusz Banachiewicz, a Polish astronomer, studied...
 - a) the Moon's motion and eclipses
 - b) volcanoes
 - c) dinosaurs
 - d) the weather
8. 🌿 In 1925, Cecilia Payne discovered that stars are made mainly of...
 - a) carbon
 - b) iron
 - c) water
 - d) hydrogen and helium
9. ☆ In 1938, Orson Welles's radio drama about a Martian invasion ("The War of the Worlds")...
 - a) was filmed in color
 - b) caused confusion among some listeners
 - c) was broadcast from the Moon
 - d) lasted ten days

MATHEMATICS



8. Era of the World Wars

1900–1950 AD

Let's travel back to the era of the two world wars, a golden age of Polish mathematics. In 1915 Wacław Sierpiński described a figure made of triangles that we now call the Sierpiński triangle. Hugo Steinhaus applied mathematics to fields such as medicine and nature. Stefan Banach created Banach spaces, a concept from functional analysis, and with his colleagues he wrote down problems in a café in Lwów in the "Scottish Book." His student Stanisław Ulam later worked in the US on the Manhattan Project. Srinivasa Ramanujan, a self-taught mathematician from India, wrote down thousands of formulas in notebooks. Emmy Noether proved that conservation laws follow from symmetries of nature. In 1931 Kurt Gödel showed that every sufficiently rich system of mathematics has statements that can be neither proved nor disproved. And in the 1930s Joseph Schillinger developed a system of composing music based on mathematics. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In the 1930s, Joseph Schillinger developed a system of composing music based on...
 - a) colors
 - b) smells
 - c) horoscopes
 - d) mathematics
2. 👁️ Stefan Banach and his colleagues wrote down math problems in a café in Lwów in the...
 - a) "Book of Records"
 - b) "Jungle Book"
 - c) "Scottish Book"
 - d) "Book of Numbers"
3. 📖 Banach spaces are a concept from...
 - a) poetry
 - b) functional analysis
 - c) astronomy
 - d) biology
4. 🧮 In 1931, Kurt Gödel proved that in every sufficiently rich system of mathematics there are statements that...
 - a) cannot be written down
 - b) are always false
 - c) can be neither proved nor disproved
 - d) no one will ever read
5. ⚡ In 1915, Wacław Sierpiński described a figure made of triangles, which today we call the...
 - a) Rubik's Cube
 - b) Sierpiński triangle
 - c) Pascal triangle
 - d) Sierpiński circle
6. 🧑 Stanisław Ulam, a student of Banach, worked in the US on...
 - a) building a bridge
 - b) the Manhattan Project
 - c) coding games
 - d) only Moon flights
7. 💜 Srinivasa Ramanujan, a self-taught mathematician from India, wrote down thousands of formulas...
 - a) in notebooks
 - b) in the sand
 - c) on walls
 - d) on a phone
8. 🌿 Polish mathematician Hugo Steinhaus applied mathematics to fields such as...
 - a) cooking
 - b) soccer
 - c) medicine and nature
 - d) fashion
9. ☆ Emmy Noether proved that conservation laws (e.g. of energy) follow from...
 - a) the will of the gods
 - b) market laws
 - c) symmetries of nature
 - d) chance

OTHER (TECHNOLOGY)



8. Era of the World Wars

1900–1950 AD

Let's travel back to the era of the two world wars, full of inventions that entered everyday life. In 1908 Kazimierz Prószyński built a handheld film camera called the Aeroscope. In the USA in 1916 the first self-service stores opened. In the 1920s the zipper became popular because it was faster than buttons, and Clarence Birdseye popularized frozen food after watching Inuit people freezing fish. In the 1930s the magnetic tape recorder was developed in Germany. Kodachrome, from 1935, was the first color photographic film for amateurs. In 1938 László Bíró patented the ballpoint pen. Wartime inventions such as radar later made their way into homes in the form of the microwave oven. And in 1948 the Polaroid camera appeared, which developed a photo instantly. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎧 **The magnetic tape recorder was developed in the 1930s in...**
 - a) Germany
 - b) Poland
 - c) China
 - d) Brazil
2. 👁 **The first color photographic film for amateurs was...**
 - a) Kodachrome (1935)
 - b) Kodak Black
 - c) Polaroid (2000)
 - d) Instagram
3. 🖋 **The ballpoint pen was patented in 1938 by...**
 - a) Thomas Edison
 - b) Walt Disney
 - c) Henry Ford
 - d) László Bíró
4. ➡ **The zipper became popular in the 1920s because it was...**
 - a) prettier than pockets
 - b) faster than buttons
 - c) cheaper than shoes
 - d) quieter than Velcro
5. 🎥 **In 1908, Kazimierz Prószyński built a handheld film camera called the...**
 - a) Aeroscope
 - b) iPhone
 - c) projector
 - d) television
6. 🛒 **The first self-service stores, where customers take goods from the shelves themselves, opened in...**
 - a) the USA in 1916
 - b) China in 2000
 - c) Rome
 - d) Poland in 1850
7. 💖 **The Polaroid camera, which developed a photo instantly, appeared in...**
 - a) 1970
 - b) 1990
 - c) 1900
 - d) 1948
8. 🍷 **Frozen food was popularized in the 1920s by Clarence Birdseye, who got the idea from watching...**
 - a) penguins
 - b) Inuit people freezing fish
 - c) the Romans
 - d) bees
9. ⭐ **Wartime inventions such as radar later made their way into homes in the form of the...**
 - a) color television
 - b) electric iron
 - c) microwave oven
 - d) refrigerator

1.a.2.a.3.d.4.b.5.a.6.a.7.d.8.b.9.c

ANSWERS

BIOLOGY

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when biology looked inside the cell. In 1900 three researchers confirmed Gregor Mendel's laws of heredity, rediscovering him. In 1909 Wilhelm Johannsen introduced the word "gene." Thomas Morgan, studying fruit flies, discovered that genes are located on chromosomes. In 1944 Oswald Avery showed that hereditary information is carried by DNA. Karl von Frisch discovered that bees tell each other where food is by dancing. In 1936 the last thylacine died in a zoo, and in 1938 the coelacanth, a "living fossil," was discovered off the coast of South Africa. After World War I fewer than 60 European bison were left, and only in zoos, but in 1929 the first captive-bred bison returned to the Białowieża Forest. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Karl von Frisch discovered that bees tell each other where food is by...**
 - a) singing
 - b) writing
 - c) dancing
 - d) the smell of colors
2. 👁️ **Thomas Morgan studied heredity in fruit flies and discovered that genes are located on...**
 - a) leaves
 - b) chromosomes
 - c) bones
 - d) wings
3. 📖 **The word "gene" was introduced in 1909 by...**
 - a) Louis Pasteur
 - b) Wilhelm Johannsen
 - c) Gregor Mendel
 - d) Charles Darwin
4. ➡️ **In 1944, Oswald Avery showed that hereditary information is carried by...**
 - a) DNA
 - b) sugar
 - c) fat
 - d) protein
5. 🐅 **The last thylacine (Tasmanian tiger) died in a zoo in...**
 - a) 1986
 - b) 1836
 - c) 1936
 - d) 2006
6. 🐉 **The coelacanth, a "living fossil," was discovered in 1938 off the coast of...**
 - a) Norway
 - b) Poland
 - c) Japan
 - d) South Africa
7. 💖 **Gregor Mendel was "rediscovered" in 1900 when three researchers...**
 - a) found his grave
 - b) confirmed his laws of heredity
 - c) translated his poems
 - d) built him a monument
8. 🌿 **The first captive-bred European bison returned to the Białowieża Forest in...**
 - a) 1929
 - b) 1829
 - c) 2009
 - d) 1989
9. ⭐ **After World War I, fewer than 60 European bison were left in the world, and only in...**
 - a) zoos
 - b) Siberian forests
 - c) Africa
 - d) America

CHEMISTRY

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when chemistry revealed the secrets of matter. In 1909 Søren Sørensen introduced the pH scale, which measures acidity; distilled water has a pH of 7, which is neutral. The Polish biochemist Kazimierz Funk coined the word "vitamine." In 1911 Marie Skłodowska-Curie received her second Nobel Prize, this time in chemistry. In 1932 the Radium Institute in Warsaw opened on the initiative of Marie Skłodowska-Curie. In 1935 her daughter Irène Joliot-Curie and her husband won the Nobel Prize for artificial radioactivity. In the 1930s Linus Pauling studied chemical bonds. In 1938 Otto Hahn and Lise Meitner discovered nuclear fission. And in 1944 element 96, curium, was discovered and named in honor of Marie and Pierre Curie. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏆 In 1911, Marie Skłodowska-Curie received her second Nobel Prize, this time in...
 - a) medicine
 - b) physics
 - c) chemistry
 - d) literature
 2. 👁 In the 1930s, Linus Pauling studied...
 - a) dinosaurs
 - b) chemical bonds
 - c) volcanoes
 - d) stars
 3. 🗨 The pH scale, which measures acidity, was introduced in 1909 by...
 - a) Mendeleev
 - b) Marie Curie
 - c) Søren Sørensen
 - d) Alfred Nobel
 4. 🧪 What is the pH of distilled water?
-
5. 🏆 In 1935, Irène Joliot-Curie and her husband received the Nobel Prize for...
 - a) a vaccine
 - b) artificial radioactivity
 - c) discovering oxygen
 - d) inventing plastic
 6. 👤 The Radium Institute in Warsaw opened in 1932 on the initiative of...
 - a) Marie Skłodowska-Curie
 - b) Józef Piłsudski
 - c) Jan Czochoński
 - d) Ignacy Mościcki
 7. ❤ In 1938, Otto Hahn and Lise Meitner discovered...
 - a) oxygen
 - b) hydrogen
 - c) bacteria
 - d) nuclear fission
 8. 🌿 Vitamins were discovered in the early 20th century. The Polish biochemist who coined the word "vitamine" was...
 - a) Rudolf Weigl
 - b) Jan Czochoński
 - c) Kazimierz Funk
 - d) Ludwik Hirszfeld
 9. ☆ Element 96, curium, was named in honor of...
 - a) Einstein
 - b) Marie and Pierre Curie
 - c) Nobel
 - d) Copernicus

PHYSICS

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when physics looked deep inside the atom. In 1900 Max Planck proposed that energy is emitted in packets called quanta. In 1905, in his special theory of relativity, Albert Einstein showed that nothing moves faster than light. In his free time Einstein played the violin. The Polish physicist Marian Smoluchowski, independently of Einstein, explained the Brownian motion of particles in a liquid. In 1911 Ernest Rutherford discovered that most of an atom is empty space around a tiny nucleus. In 1912, during balloon flights, Victor Hess discovered cosmic rays. In 1913 Niels Bohr's model showed electrons circling the nucleus in fixed orbits. In 1927 Werner Heisenberg stated that one cannot know precisely at the same time a particle's position and momentum. And in 1939 Einstein signed a letter warning that Germany might build an atomic bomb. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎸 In his free time, Albert Einstein played the...
 - a) guitar
 - b) trumpet
 - c) violin
 - d) drums
2. 👁️ Niels Bohr's model of the atom (1913) showed electrons circling the nucleus in...
 - a) straight lines
 - b) fixed orbits
 - c) squares
 - d) inward spirals
3. 📄 In 1900, Max Planck proposed that energy is emitted in packets called...
 - a) quanta
 - b) waves
 - c) atoms
 - d) grams
4. ⚡ In 1911, Ernest Rutherford discovered that most of an atom is...
 - a) empty space around a tiny nucleus
 - b) a hard ball
 - c) water
 - d) fire
5. 🚀 In his special theory of relativity (1905), Einstein showed that nothing moves faster than...
 - a) sound
 - b) a rocket
 - c) the wind
 - d) light
6. 🧑 Polish physicist Marian Smoluchowski explained (independently of Einstein)...
 - a) how radio waves travel through air
 - b) the internal structure of stars
 - c) the Brownian motion of particles in a liquid
 - d) the nature of X-rays
7. 💖 Werner Heisenberg formulated the principle that one cannot know precisely at the same time a particle's...
 - a) position and momentum
 - b) color and smell
 - c) name and weight
 - d) mass and age
8. 🌿 In 1912, during balloon flights, Victor Hess discovered...
 - a) nighttime rainbows
 - b) electricity in clouds
 - c) the ozone hole
 - d) cosmic rays
9. ⭐ In 1939, Einstein signed a letter to the US president warning that Germany might build...
 - a) radar
 - b) a computer
 - c) an atomic bomb
 - d) a rocket to the Moon

ENVIRONMENT

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when people reached the ends of the Earth and learned to protect nature. In 1902 the eruption of Mount Pelée on Martinique destroyed the town of Saint-Pierre. Roald Amundsen had learned earlier from the Inuit, so in 1911 he was the first to reach the South Pole. After losing his ship "Endurance" in 1915, Ernest Shackleton and his companions sailed 1,300 km across the ocean in a lifeboat. In Poland the nature conservation movement was developed by Władysław Szafer, who founded the State Council for Nature Conservation, and in 1932 the Białowieża National Park was established. The Dust Bowl storms in the US showed that excessive plowing destroys soil. Bombing and fires in World War II also destroyed forests and farmland. And in 1949 Aldo Leopold wrote about a land ethic and respect for nature. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 In 1911, Roald Amundsen was the first to reach...
 - a) the South Pole
 - b) the summit of Everest
 - c) the North Pole
 - d) the bottom of the ocean
2. 🇵🇱 Poland's oldest national park, Białowieża, was established as a national park in...
 - a) 1932
 - b) 1990
 - c) 2000
 - d) 1832
3. 🗨️ The nature conservation movement in Poland was developed by Władysław Szafer, who founded...
 - a) a famous Kraków soccer club
 - b) the State Council for Nature Conservation
 - c) a national sports newspaper
 - d) a puppet theater for children
4. 🌪️ The Dust Bowl storms in the US showed that...
 - a) wind doesn't matter
 - b) grain grows without soil
 - c) rain is harmful
 - d) excessive plowing destroys soil
5. 🚢 After losing his ship "Endurance" (1915), Ernest Shackleton and his companions sailed...
 - a) home by plane
 - b) 1,300 km across the ocean in a lifeboat
 - c) in a submarine
 - d) across Antarctica by swimming
6. 🧑‍🚀 Roald Amundsen survived in the Arctic and reached the Pole because he had learned earlier from the...
 - a) Egyptians
 - b) Romans
 - c) Inuit
 - d) Vikings in a museum
7. 💖 In "A Sand County Almanac" (1949), Aldo Leopold wrote about...
 - a) gold mines
 - b) war
 - c) building machines
 - d) a land ethic and respect for nature
8. 🌋 The eruption of Mount Pelée (1902) on Martinique destroyed the town of...
 - a) Pompeii
 - b) Naples
 - c) Reykjavik
 - d) Saint-Pierre
9. ⭐ Bombing and fires during World War II also destroyed...
 - a) the oceans completely
 - b) glaciers
 - c) the Moon
 - d) forests and farmland

HEALTH

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when people's health was put to a great test. From 1918 to 1920 the "Spanish Lady" raged – a pandemic of influenza. In 1921 Banting and Best discovered insulin, and Calmette and Guérin introduced the BCG vaccine against tuberculosis. In the 1930s vitamin C was isolated; a lack of it causes scurvy in sailors and polar explorers. In Lwów Rudolf Weigl produced a typhus vaccine by feeding human blood to lice. His vaccine, smuggled in secretly, saved lives in ghettos and camps. During the war ration cards were meant to share scarce food fairly, and singing together in shelters helped people control their fear. After the war, in 1946, UNICEF was created to help children in need. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Singing together in shelters during bombing raids helped people...**
 - a) control their fear
 - b) see better
 - c) go without food
 - d) run faster
2. 👁️ **Food ration cards were introduced during the war in order to...**
 - a) buy tickets
 - b) collect autographs
 - c) entertain children
 - d) share scarce food fairly
3. 🗨️ **The pandemic nicknamed the "Spanish Lady" was a pandemic of...**
 - a) influenza (1918–1920)
 - b) plague
 - c) cholera
 - d) smallpox
4. 🧬 **The BCG vaccine against tuberculosis was introduced in 1921. It was developed by...**
 - a) Salk and Sabin
 - b) Fleming and Florey
 - c) Calmette and Guérin
 - d) Pasteur and Koch
5. 🦋 **In Lwów, Rudolf Weigl produced a typhus vaccine by feeding human blood to...**
 - a) leeches
 - b) lice
 - c) mice
 - d) mosquitoes
6. 🧑 **Weigl's vaccine saved lives, among other places, in...**
 - a) ghettos and camps, smuggled in secretly
 - b) ancient Rome
 - c) outer space
 - d) colonial America
7. 💖 **Insulin, which saves people with diabetes, was discovered in 1921 by...**
 - a) Banting and Best
 - b) Watson and Crick
 - c) Pasteur and Koch
 - d) Curie and Curie
8. 🍃 **Scurvy, a disease affecting sailors and polar explorers, is caused by a lack of...**
 - a) calcium
 - b) vitamin C
 - c) water
 - d) salt
9. ☆ **After the war, UNICEF (1946) was created to help...**
 - a) children in need
 - b) soldiers
 - c) factories
 - d) kings

1.a.2.d.3.a.4.c.5.b.6.a.7.a.8.b.9.a

A N S W E R S

PHYSICAL CULTURE AND SPORT

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when sport became a great festival of nations. In 1920, at the Games in Antwerp, the Olympic flag first flew, with five rings that symbolize five continents. In 1924 the first Winter Olympic Games were held in Chamonix. In 1928 in Amsterdam Halina Konopacka won Poland's first Olympic gold medal in the discus throw. In 1930 the first FIFA World Cup was played in Uruguay. In 1932 Janusz Kusociński won the 10,000 m, and Stanisława Walasiewicz won the 100 m. In 1936 in Berlin Jesse Owens won four gold medals. The Games of 1940 and 1944 were canceled because of World War II, and the skier and mountaineer Bronisław Czech died in Auschwitz. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏁 **The Olympic flag with five rings first flew at the Games in...**
 - a) Rome 1960
 - b) London 1948
 - c) Antwerp 1920
 - d) Athens 1896
2. 👁️ **The five Olympic rings symbolize...**
 - a) five sports
 - b) five victories
 - c) five cities
 - d) five continents
3. 🏆 **The first FIFA World Cup was held in 1930 in...**
 - a) Brazil
 - b) Poland
 - c) England
 - d) Uruguay
4. 🏊 **At the 1936 Berlin Olympics, Jesse Owens won...**
 - a) no medals
 - b) ten medals
 - c) four gold medals
 - d) one silver medal
5. 🏆 **In which event did Halina Konopacka win Poland's first Olympic gold medal (Amsterdam 1928)?**
 - a) 100 m sprint
 - b) Discus throw
 - c) High jump
 - d) Shot put
6. 🏃 **In 1932, Stanisława Walasiewicz (Stella Walsh) won gold in the...**
 - a) marathon
 - b) 10 km
 - c) 100 m
 - d) long jump
7. ❤️ **During the war, Bronisław Czech, a skier and mountaineer...**
 - a) ran a ski factory
 - b) fled to Switzerland
 - c) died in Auschwitz
 - d) moved to the USA
8. 🍃 **The first Winter Olympic Games were held in 1924 in...**
 - a) Innsbruck
 - b) Chamonix
 - c) Oslo
 - d) Zakopane
9. ⭐ **The Olympic Games of 1940 and 1944 were canceled because of...**
 - a) World War II
 - b) a lack of athletes
 - c) a flu epidemic
 - d) an economic crisis

MEDICINE

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when medicine gained a new weapon. In 1928 Alexander Fleming noticed that a mold had killed bacteria on a dish, and so he discovered penicillin – the first antibiotic to be mass-produced. The word "antibiotic" literally means against life, the life of bacteria. Antibiotics don't work against the flu, however, because it is caused by viruses. In 1943 streptomycin, the first effective drug against tuberculosis, was discovered in soil bacteria. During the war wounded soldiers received blood thanks to blood banks, and Red Cross doctors helped the wounded on both sides. In the Kraków Ghetto Tadeusz Pankiewicz ran a pharmacy, helping Jews. And in 1947 the Nuremberg Code established that medical research on humans requires the participant's voluntary consent. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Alexander Fleming discovered penicillin in 1928 when he noticed that a mold...**
 - a) changed the water's color
 - b) had started to glow
 - c) had killed bacteria on a dish
 - d) played like music
2. 👁 **The first antibiotic to be mass-produced was...**
 - a) aspirin
 - b) morphine
 - c) insulin
 - d) penicillin
3. 🗨 **The word "antibiotic" literally means...**
 - a) for pain
 - b) from a fungus
 - c) for health
 - d) against life (of bacteria)
4. ⚡ **Antibiotics don't work against the flu because the flu is caused by...**
 - a) viruses
 - b) fungi
 - c) parasites
 - d) bacteria
5. ⚡ **During the war, wounded soldiers received blood thanks to...**
 - a) synthetic blood
 - b) blood machines
 - c) animal blood
 - d) blood banks
6. 🧑 **During wars, Red Cross doctors...**
 - a) fought with weapons
 - b) stayed away from the front
 - c) helped the wounded on both sides
 - d) treated only their own side
7. ❤ **In the Kraków Ghetto, Tadeusz Pankiewicz ran...**
 - a) a dance school
 - b) a pharmacy, helping Jews
 - c) a factory
 - d) a restaurant
8. 🌿 **Streptomycin, the first effective drug against tuberculosis, was discovered in 1943 in...**
 - a) seawater
 - b) tea leaves
 - c) potatoes
 - d) soil bacteria
9. ⭐ **The Nuremberg Code (1947) established that medical research on humans requires...**
 - a) a fee
 - b) the participant's voluntary consent
 - c) the king's consent
 - d) hiding it from the patient

PHARMACY

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when pharmacies turned into a modern industry. Tablets began to be mass-produced using tablet press machines, and companies advertised medicines on the radio and in newspapers. In the 1930s Tadeus Reichstein and others discovered adrenal cortex hormones, such as cortisone. In 1935 the first sulfonamide drugs, like Prontosil, worked against bacteria. In 1940 Howard Florey and Ernst Chain showed that penicillin cures infections in humans, and during the war the mold was grown in huge vats. Quinine was very valuable in the war in Asia because soldiers fell ill in the tropics. Under the occupation, medicines in Poland were obtained through underground pharmacies and smuggling. After the war stricter drug testing was introduced because some drugs turned out to be harmful. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 In the 20th century, pharmaceutical companies advertised medicines on...
 - a) color TV
 - b) the radio and in newspapers
 - c) the internet
 - d) smartphones
2. 📺 Tablets began to be mass-produced using...
 - a) wine presses
 - b) printers
 - c) pharmacists' hands only
 - d) tablet press machines
3. 🗨️ The first sulfonamide drugs (Prontosil, 1935) worked against...
 - a) cancer
 - b) viruses
 - c) headaches only
 - d) bacteria
4. ⚗️ In 1940, Howard Florey and Ernst Chain showed that penicillin...
 - a) is a poison
 - b) dyes hair
 - c) does nothing
 - d) cures infections in humans
5. 🌱 To produce large amounts of penicillin during the war, the mold was grown in...
 - a) refrigerators
 - b) aquariums
 - c) gardens
 - d) huge vats
6. 🧑 During the occupation, medicines in Poland were often obtained through...
 - a) online stores
 - b) underground pharmacies and smuggling
 - c) free pharmacies
 - d) help from the Moon
7. ❤️ In the 1930s, Tadeus Reichstein and others discovered adrenal cortex hormones, such as...
 - a) adrenaline
 - b) cortisone
 - c) insulin
 - d) vitamin C
8. 🌿 Quinine, a malaria drug, was very valuable in the war in Asia because...
 - a) it was sweet
 - b) soldiers fell ill in the tropics
 - c) it gave strength
 - d) it healed wounds
9. ☆ After the war, stricter drug testing was introduced because...
 - a) some drugs turned out to be harmful
 - b) drugs were too cheap
 - c) people stopped getting sick
 - d) pharmacies were banned

AGRICULTURE

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when farming began to be mechanized. In 1917 the Fordson tractor appeared, one of the first cheap, mass-produced tractors. A combine harvester combines reaping and threshing. Artificial fertilizers from the Haber–Bosch process made it possible to feed many more people. In the Second Polish Republic land reform meant dividing large estates for peasants, and the dożynki was a great harvest celebration with wreaths. In 1932–1933 the Great Famine in Ukraine, called the Holodomor, was caused by the policies of the Soviet authorities. Nikolai Vavilov assembled the largest collection of crop plant seeds. During the war allotment gardens and "victory gardens" helped people get more food. And in 1944 Norman Borlaug began his work in Mexico, from which the "Green Revolution" grew. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Between the wars, the dożynki (harvest festival) in Poland was...**
 - a) a factory holiday
 - b) celebrated in winter
 - c) banned
 - d) a great harvest celebration with wreaths
2. 👁️ **The Fordson tractor (1917) was one of the first...**
 - a) sports cars
 - b) tanks
 - c) cheap, mass-produced tractors
 - d) crop-dusting planes
3. 🗨️ **The "Green Revolution" in agriculture began with Norman Borlaug's work in...**
 - a) Australia
 - b) Mexico (from 1944)
 - c) Poland
 - d) Egypt
4. ⚗️ **Artificial fertilizers from the Haber–Bosch process made it possible to...**
 - a) farm on ice
 - b) eliminate rain
 - c) reduce crop yields
 - d) feed many more people
5. 🌾 **A combine harvester combines...**
 - a) sowing and plowing
 - b) planting trees
 - c) watering and fertilizing
 - d) reaping and threshing
6. 🏘️ **Land reform in the Second Polish Republic involved...**
 - a) banning crops
 - b) closing villages
 - c) dividing large estates for peasants
 - d) building railroads
7. ❤️ **The Great Famine in Ukraine (1932–1933), called the Holodomor, was caused by...**
 - a) a war with Poland
 - b) the policies of the Soviet authorities
 - c) a volcanic eruption
 - d) a worldwide drought
8. 🌿 **Nikolai Vavilov assembled the largest collection of...**
 - a) crop plant seeds
 - b) insects
 - c) stones
 - d) birds
9. ⭐ **During the war, allotment gardens and "victory gardens" helped people...**
 - a) raise horses
 - b) get more food
 - c) build shelters
 - d) hide weapons

VETERINARY SCIENCE

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when animals served side by side with soldiers. In World War I horses pulled guns and supply wagons, medical dogs found wounded soldiers, and homing pigeons were valuable because they return to their loft from far away. Rin Tin Tin, a dog rescued from the front, became a Hollywood film star. In Japan the dog Hachikō waited at a station for his late owner for about 10 years. In 1939 the Polish cavalry fought mostly on foot, and horses were used for transport. The British Dickin Medal was awarded to animals for bravery in war. Veterinary vaccines, for example against rabies in dogs, protected both animals and people. And in the 1940s rockets carried fruit flies and monkeys, long before Laika flew into space in 1957. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **In World War I, horses...**
 - a) flew airplanes
 - b) pulled guns and supply wagons
 - c) took no part
 - d) sailed boats
2. 🐶 **Rin Tin Tin, a dog rescued from the World War I front, became...**
 - a) the first dog in space
 - b) the king's mascot
 - c) a police dog
 - d) a Hollywood film star
3. 📖 **Medical dogs in World War I...**
 - a) wrote letters
 - b) cooked soup
 - c) drove tanks
 - d) found wounded soldiers
4. 🕊 **Homing pigeons were valuable in wartime because they...**
 - a) carry heavy packages
 - b) return to their loft from far away
 - c) fly only at night
 - d) can talk
5. 🚩 **In 1939, the Polish cavalry...**
 - a) rode elephants
 - b) charged tanks with sabers
 - c) fought mostly on foot
 - d) did not exist
6. 🏆 **The British Dickin Medal was awarded to...**
 - a) animals for bravery in war
 - b) doctors
 - c) pilots
 - d) children
7. ❤ **For about 10 years, Hachikō, a dog from Japan...**
 - a) rescued drowning people
 - b) guarded the emperor
 - c) waited at a station for his late owner
 - d) drove a train
8. 🌿 **Veterinary vaccines, e.g. against rabies in dogs, helped protect...**
 - a) only cows
 - b) machines
 - c) both animals and people
 - d) only fish
9. ⭐ **The dog Laika went to space later (1957), but already in the 1940s rockets carried...**
 - a) elephants
 - b) fruit flies and monkeys
 - c) horses
 - d) aquarium fish



ANIMAL SCIENCE AND FISHERIES

8. Era of the World Wars

1900–1950 AD

Let's travel back to the era of the two world wars, when animal and fish farming expanded. Shepherds in the Alps and the Tatras hung bells on their cows and sheep. The hardy Hutsul horse comes from the Eastern Carpathians, and in the 1930s Tadeusz Vetulani bred back the Polish konik, a descendant of wild tarpans. In Poland the most famous carp ponds lie near Zator. After 1920 Polish sea fishing developed thanks to access to the Baltic. In the 1930s artificial insemination was developed in the USSR and the US to improve cattle breeds faster. Raising laying hens on a large scale meant that eggs became cheap. Even then people noticed overfishing, when more fish are caught than are born. And during the war hunger meant that many farm animals were eaten or requisitioned. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Shepherds in the Alps and the Tatras hung on their cows and sheep...**
 - a) rings
 - b) flags
 - c) lamps
 - d) bells
2. 👁 **The Hutsul horse breed comes from the...**
 - a) Arabian Peninsula
 - b) English countryside
 - c) Eastern Carpathians
 - d) Mongolian steppe
3. 🐟 **Carp farming in Poland has a long tradition. The most famous ponds are near...**
 - a) Gdańsk
 - b) Zator
 - c) Zakopane
 - d) Suwałki
4. ⚡ **Artificial insemination of animals was developed in the 1930s in the USSR and the US in order to...**
 - a) reduce the number of cows
 - b) raise fish
 - c) improve cattle breeds faster
 - d) ban animal breeding
5. 🌊 **Polish sea fishing developed after 1920 thanks to...**
 - a) mountain rivers
 - b) access to the Baltic
 - c) city lakes
 - d) swimming pools
6. 🐔 **In the 20th century, laying hens began to be raised on a large scale, which meant that...**
 - a) hens learned to fly
 - b) hens stopped laying
 - c) eggs disappeared
 - d) eggs became cheap
7. 💜 **The Polish konik, a descendant of wild tarpans, was bred back in the 1930s by...**
 - a) Józef Piłsudski
 - b) Charles Darwin
 - c) Tadeusz Vetulani
 - d) Nicolaus Copernicus
8. 🌿 **Overfishing is when...**
 - a) there are too many fish in the sea
 - b) no fish are caught at all
 - c) more fish are caught than are born
 - d) the water freezes
9. ☆ **During the war, hunger meant that many farm animals were...**
 - a) set free
 - b) eaten or requisitioned
 - c) insured
 - d) sent abroad

OTHER (NATURE)

8. Era of the World Wars

1900–1950 AD



Let's travel back to the era of the two world wars, when explorers reached the farthest corners of the Earth. In 1909 Robert Peary announced that he had reached the North Pole. In 1912 Alfred Wegener presented the theory of continental drift, because he noticed that the coasts of Africa and South America fit together like puzzle pieces. Wegener died in 1930 during an expedition to Greenland. The Polish geologist Antoni Bolesław Dobrowolski studied the glaciers of Antarctica. In 1924 George Mallory and Andrew Irvine attempted to climb Everest. In the 1930s nature sound recorders made it possible to record birdsong. In 1943 Jacques Cousteau co-created the Aqua-Lung for diving, and in 1947 Thor Heyerdahl crossed the Pacific on a raft called Kon-Tiki. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎧 In the 1930s, nature sound recorders made it possible to record...
 - a) sounds from Mars
 - b) animals' thoughts
 - c) dinosaur calls
 - d) birdsong
2. 🏔 Everest was not climbed until 1953, but in 1924 an attempt was made by George Mallory and...
 - a) Tenzing Norgay
 - b) Andrew Irvine
 - c) Edmund Hillary
 - d) Reinhold Messner
3. 🗨 In 1912, Alfred Wegener presented the theory of...
 - a) the Big Bang
 - b) gravity
 - c) continental drift
 - d) evolution
4. ➡ Wegener noticed that the coasts of Africa and South America...
 - a) fit together like puzzle pieces
 - b) are identical in color
 - c) have no sand
 - d) are made of ice
5. 🏔 In 1909, Robert Peary announced that he had reached...
 - a) the bottom of the Mariana Trench
 - b) Everest
 - c) the South Pole
 - d) the North Pole
6. 🚣 In 1947, Thor Heyerdahl crossed the Pacific on a raft called...
 - a) Titanic
 - b) Kon-Tiki
 - c) Nautilus
 - d) Ark
7. ❤ In 1943, Jacques Cousteau co-created...
 - a) the airplane
 - b) the Aqua-Lung for diving
 - c) the telephone
 - d) the compass
8. 🌿 Polish geologist and explorer Antoni Boleśław Dobrowolski studied...
 - a) the jungle
 - b) the glaciers of Antarctica
 - c) the Himalayas
 - d) the Sahara
9. ☆ Wegener died in 1930 during an expedition to...
 - a) the Moon
 - b) the Amazon
 - c) the Sahara
 - d) Greenland